Chargeback Prevention: The CFO Case

Cinderhaven Data Platform — Executive Summary

Executive Summary

Cinderhaven's chargebacks are not inevitable. Most trace to a handful of upstream data conditions — and those conditions are fixable before the order ships.

<div class="headline-number">$$446K</div>
<div class="headline-label">Annual chargebacks</div>

<div class="headline-number">$$325K</div>
<div class="headline-label">Preventable at source</div>

<div class="headline-number">4 causes</div>
<div class="headline-label">Fixable root conditions</div>

Grouping Cinderhaven's historical chargebacks by root cause and applying a fixed preventability fraction to each category flags **73%** of the annual total as traceable to four upstream data conditions — missing case dimensions, barcode registration gaps, late ASN filing, and pricing discrepancies. This share is an assumption-driven estimate from the roadmap below, not a model output. Each condition is detectable before the order ships. Each has a specific operational fix.

The deductions team currently works these chargebacks reactively, winning 30–45% in disputes. Prevention requires no dispute process. Fixing the data conditions eliminates the chargeback at source and recovers the full amount, not a fraction of it.

Prevention roadmap

Table 1

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#	Root cause	Annual loss	Prevention value	Fix
1	Logistics Overage	$245K	$171K	Audit pick-and-pack procedures; add case-count ver

#	Root cause	Annual loss	Prevention value	Fix
2	Data Compliance Error	$143K	$114K	Populate missing barcodes (GTIN-14, UPC) and ve
3	ASN Timing Infraction	$42K	$29K	Submit Advance Ship Notices within the retailer-re
4	Pricing Discrepancy	$17K	$10K	Reconcile PO unit prices against approved price list

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The economics

Dispute recovery returns 30–45 cents per dollar. Prevention returns 100 cents per dollar — and eliminates the working capital drag of post-ship disputes, the labor cost of the deductions team, and the carrier relationship risk from repeated chargebacks.

A companion model demonstrates the pre-ship scoring workflow. One caveat is essential to state plainly: on Cinderhaven's *real* chargeback records the upstream features carry no predictive signal (AUC   0.50 — real chargebacks are effectively random with respect to data quality). The model is therefore trained on **synthetic** labels drawn from a known formula (risk   ASN lateness × data-quality gaps), and it recovers that formula. On held-out synthetic data it flags 72% of chargebacks at a 0.5 threshold (recall) and reaches an AUC of 0.70. Predictive skill on the real target is effectively zero; the durable value here is the prevention roadmap, not the score. The question is which data conditions to fix first. This roadmap answers that by prevention value.

Produced using Cinderhaven synthetic demonstration data. Figures are illustrative of a repeatable methodology, not actual Cinderhaven financials. Real chargebacks carry no predictive signal (AUC 0.50); the model is trained on synthetic labels and reaches AUC 0.70 on held-out synthetic data.